

Evolution of Indian Batting in ODIs: A Univariate Time Series Analysis

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INTRODUCTION

Cricket is the most popular sport in India by far, and is played almost everywhere in the country. We intend to analyse the structural changes in the average annual run-rate, i.e., how many runs are scored in each over, a measure of how much bat dominates the ball or how aggressively the teams bat. The general view is that with the advent of new technology, pitches that are more batsmen-friendly and shorter boundaries, the average run rate seems to have increased. Some analysts attribute this change to the emergence of newer formats and other innovations in the game. In recent times with high scores being chased easily in all formats of the game, it is being said that this is due to the effect of T20 games on other formats of the game. We plan to model and empirically test for this effect.

DATA

The annual data for the analysis is downloaded from the **ESPN Cricinfo** website. We have considered **Run Rate Per Over (RPO) of India** for the One Day Internationals (ODIs) played in the period **1975-2019**. Although we had data from 1974 also, we have decided to delete those observations as very few (only 2) matches were played by India in this year.

As there are factors other than what we plan to study affecting the runs per over such as the weather conditions, type of teams playing, condition on the pitch, etc., an innings by innings study of the time series would not have been feasible. To nullify the effect Cricinfo has converted the data into an annual series by taking the average for an entire year. As matches are played throughout the year and in all weather conditions, we hope to nullify the effects of these factors by averaging out the data.

Minimum	Q1	Median	Q3	Mean	Maximum	Variance
3.230	4.390	4.890	5.430	4.844	5.930	0.462

Summary Statistics

Visualising the data and Preliminary tests

Runs per over plotted against the years

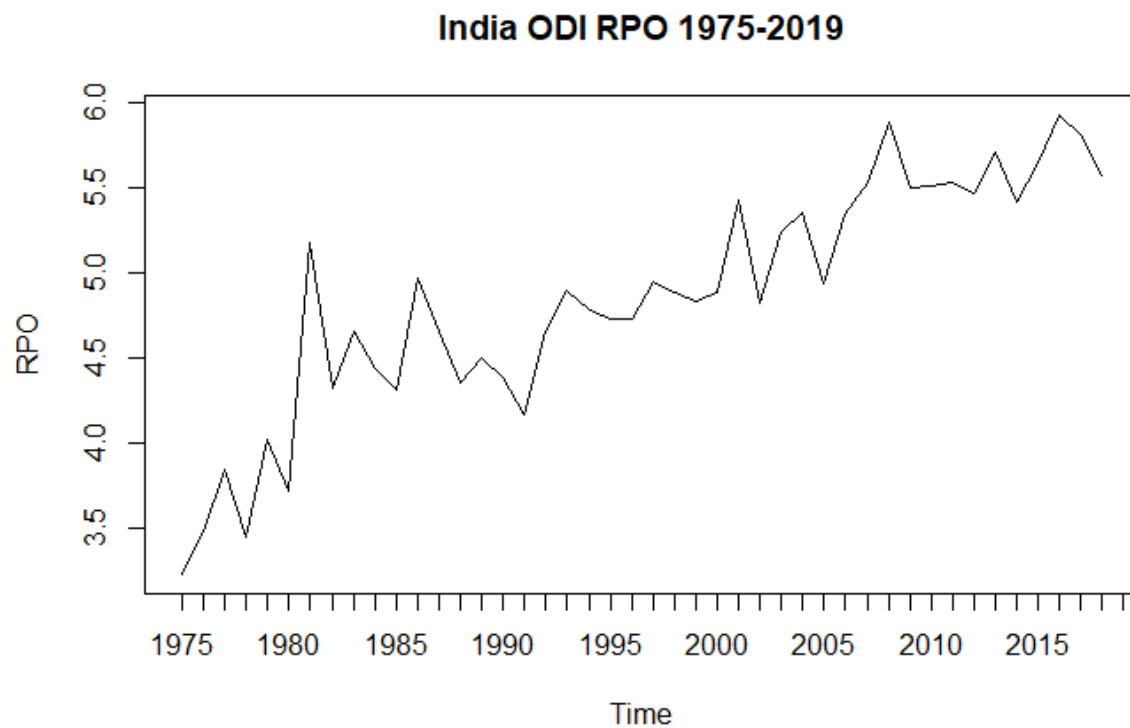


Figure 1

We observe an increasing trend in the data. To observe this trend a linear regression model is fit to the data, regressing the RPO values on years (Time).

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.56007 -0.13838 -0.00514  0.13720  1.05738

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -90.036307   7.093668  -12.69 5.78e-16 ***
Data$Year     0.047507   0.003551   13.38 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The slope is significant (p-value is almost zero). Hence we conclude trend is present in the series.

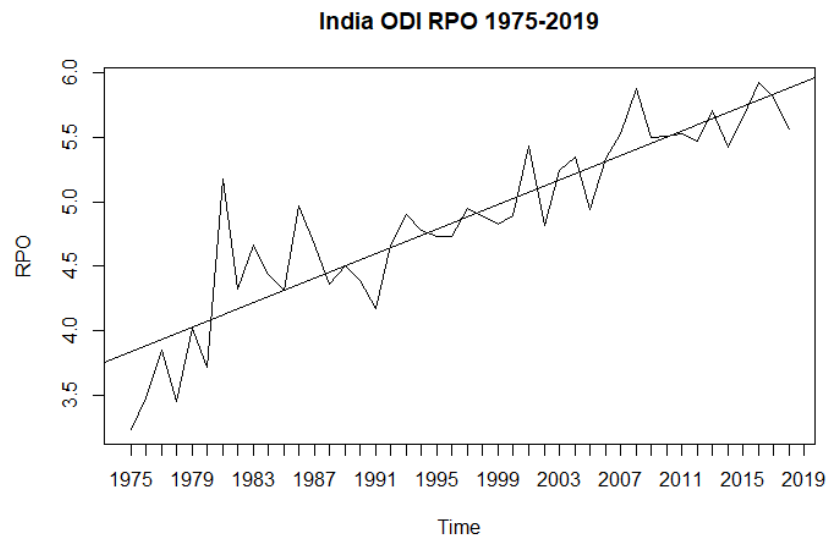


Figure 2

The histogram and QQ plot plotted as follows :

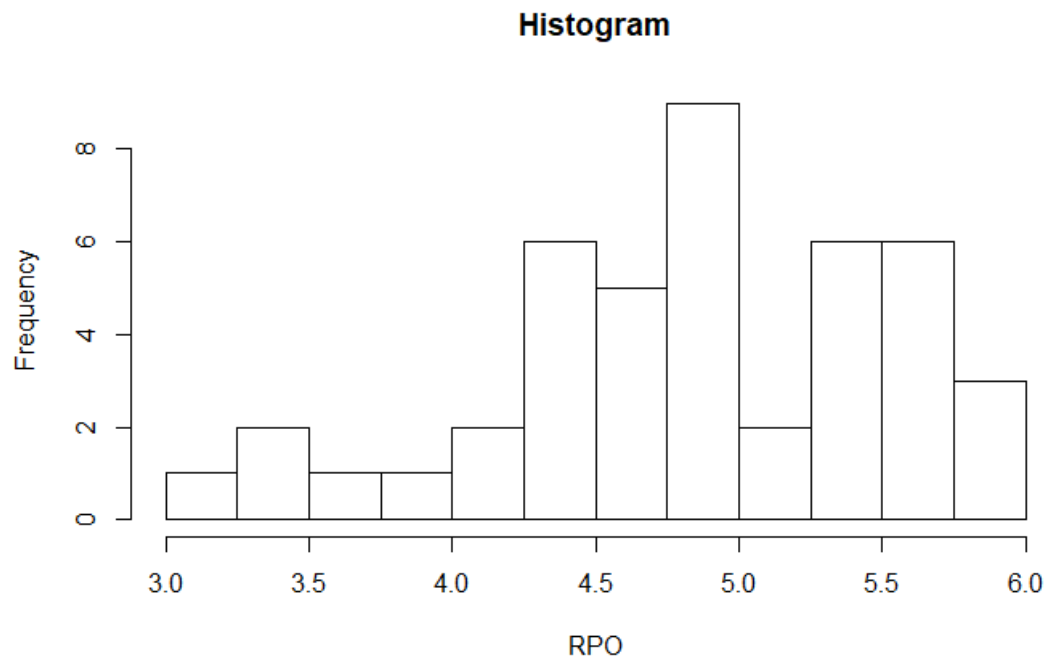


Figure 3

The graphs (Fig 3 and 4) do not show much departure from Normality. Formal tests carried out to have final conclusion.

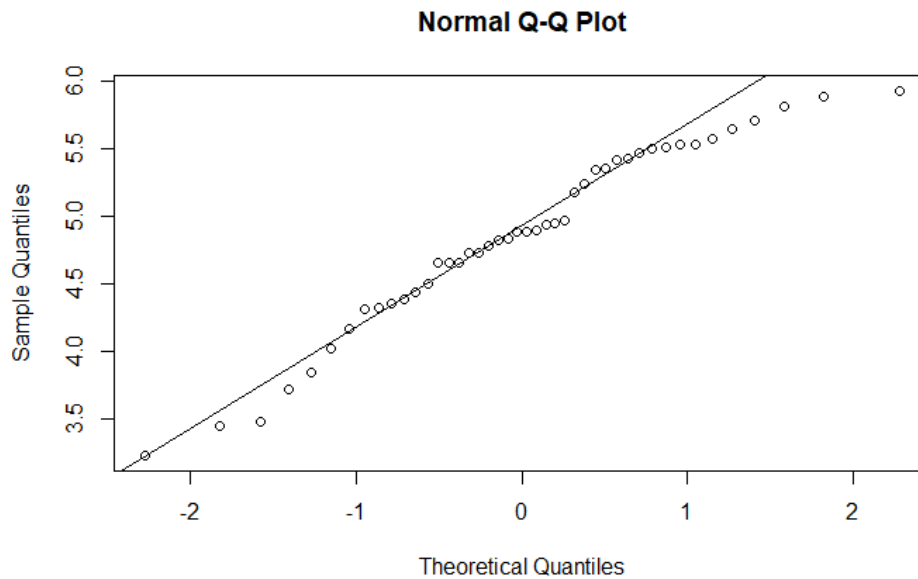


Figure 4

1) Kolmogorov Smirnov Test for Normality

The null hypothesis H_0 : The data are Normal against alternative hypothesis H_1 : The data are non - Normal are tested. The summary is as follows :

```
One-sample kolmogorov-smirnov test
data: Data_TS
D = 0.10153, p-value = 0.7548
alternative hypothesis: two-sided
```

The null hypothesis is accepted at 5% los as p value is 0.755. Hence data follow Normal distribution.

2) Anderson – Darling test

The null hypothesis H_0 : The data are Normal against alternative hypothesis H_1 : The data are non - Normal are tested. The summary is as follows :

```
Anderson-Darling normality test
data: Data_TS
A = 0.51173, p-value = 0.1852
```

The null hypothesis is accepted at 5% los as p value is 0.185. Hence data follow Normal distribution.

Hence from both the tests we conclude that data are normally distributed.

ACF & PACF of the Series

The ACF of the series is as follows :

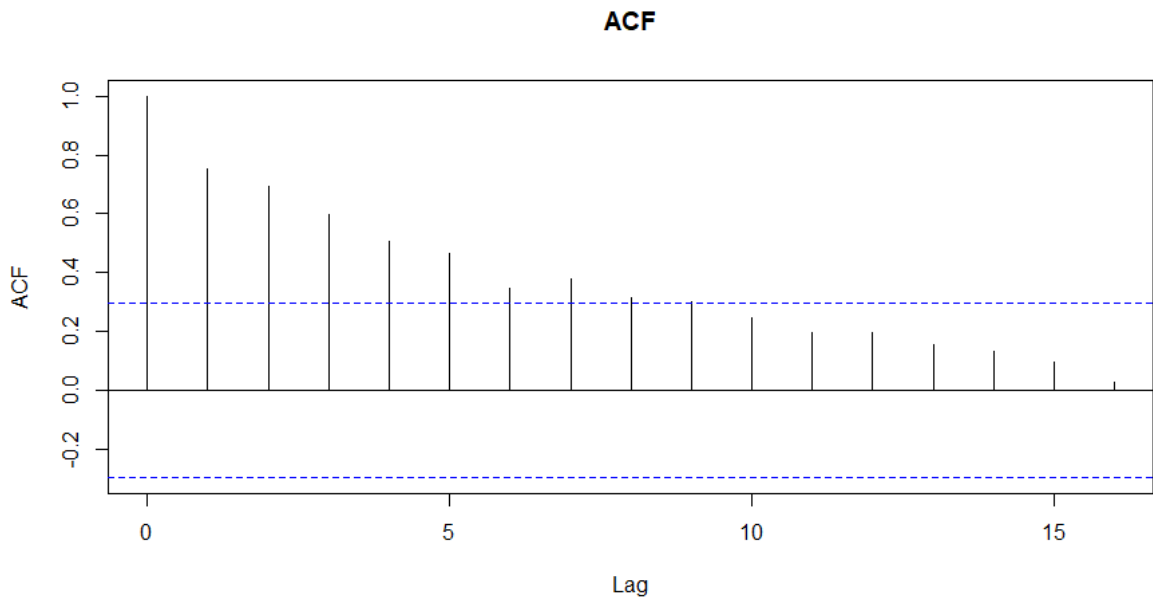


Figure 5

The decay of the ACF suggests that there is trend present in the original series.

To check whether the series is stationary, a formal test : **Dickey – Fuller Test** is carried out. The null hypothesis H_0 : The series is not stationary against H_1 : The series is stationary is tested.

```
Augmented Dickey-Fuller Test
data: Data_TS
Dickey-Fuller = -3.711, Lag order = 3, p-value = 0.03543
alternative hypothesis: stationary
```

With p value observed to be **0.035**, the null hypothesis is accepted at 1% los. Hence the series is not stationary.

Differencing of order 1 is carried out to remove the trend and resulting series is plotted.

Differencing the Series

The trend seems to be gone, but there exists one outlier at the 6th position (See Fig 6). Before further analysis is carried out, this observation is removed. This outlier corresponds to year 1982, in which India chased down small total in few overs, resulting in a higher RPO.

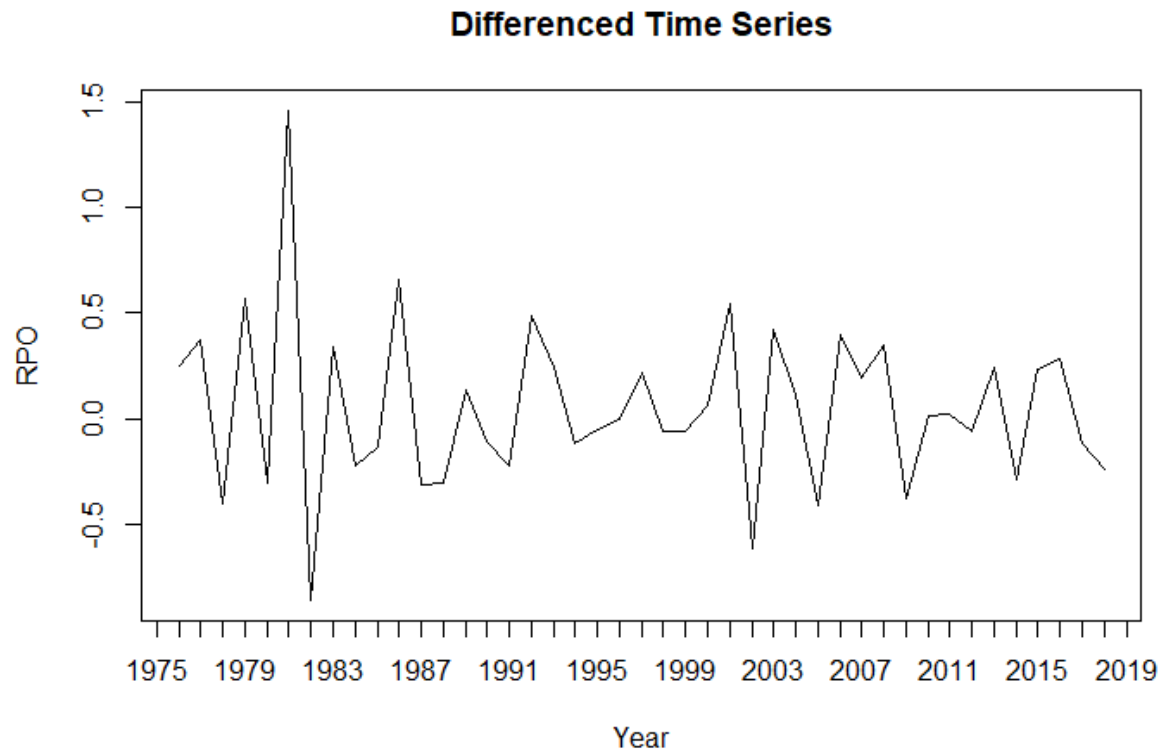


Figure 6

The plot after removing the outlier is as follows :

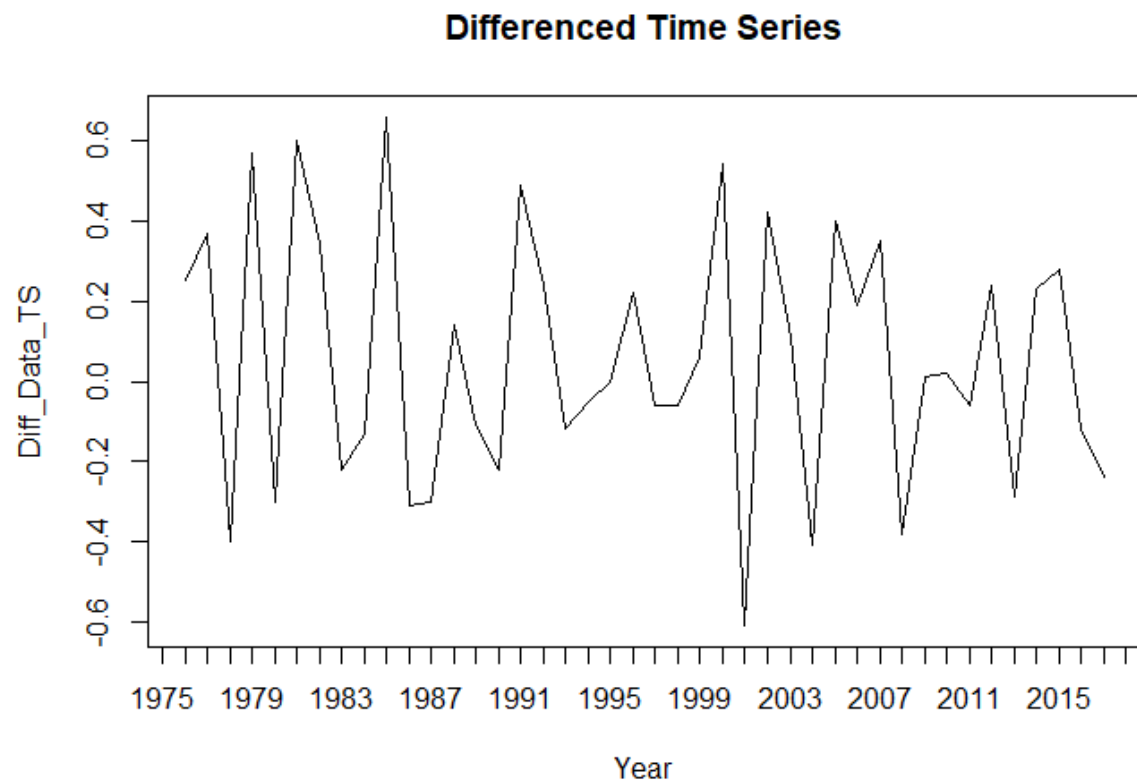


Figure 7

Dickey – Fuller test is carried out on this differenced time series again to check stationarity.

Augmented Dickey-Fuller Test

```
data: Diff_Data_TS  
Dickey-Fuller = -3.829, Lag order = 3, p-value = 0.02696  
alternative hypothesis: stationary
```

The null hypothesis is rejected at both 1% and 5% los, hence the series is stationary.

The ACF of the differenced series is plotted as follows :

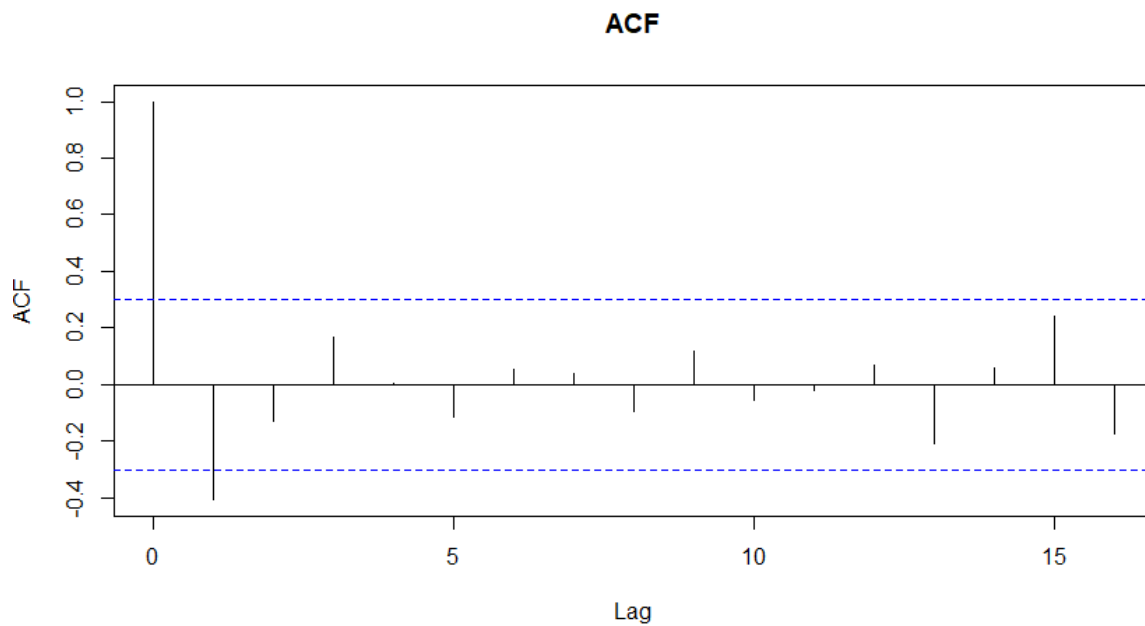


Figure 8

The PACF of the series is plotted as follows.

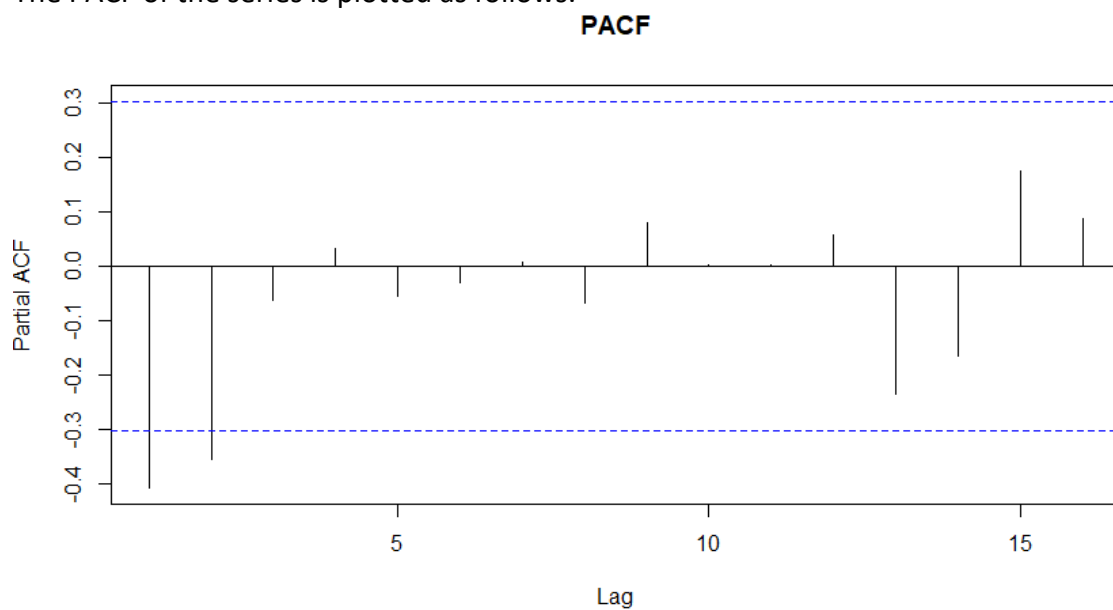


Figure 9

Two spikes observed at lag 1 and 2. Thus AR parameter can be taken as 2 initially.

Fitting the Model

Initially **ARIMA(2,1,1)** is fit to **the original series**. The summary for the model is as follows :

```

              Estimate      SE t.value p.value
ar1         -0.4439 0.3691 -1.2026  0.2366
ar2         -0.3369 0.2175 -1.5494  0.1296
ma1         -0.1773 0.3873 -0.4579  0.6497
constant     0.0563 0.0189  2.9783  0.0050

```

All the coefficients seem to be non-significant here. Thus the model can be improved further.

The model AIC , AICc and BIC are compared with few competing models.

ARIMA Order	AIC	AICc	BIC
(2, 1, 1)	0.3953226	0.4210626	0.602188
(1, 1, 1)	0.3655933	0.3806309	0.5310857
(1, 1, 2)	0.3975152	0.4232552	0.6043806
(0, 1, 1)	0.3312828	0.3386088	0.4554021
(1, 1, 0)	0.4686399	0.4759659	0.5927591
(0, 1, 0)	0.6046819	0.6070628	0.687428

Table 1

The AIC, AICc and BIC, all three are minimum for the model **ARIMA(0,1,1)**.

The summary for this model is :

```

Coefficients:
      ma1      constant
      -0.6501      0.0552
s.e.    0.1460      0.0150

sigma^2 estimated as 0.06977:  log likelihood = -3.96,  aic = 13.91

$degrees_of_freedom
[1] 40

$table
      Estimate      SE t.value p.value
ma1      -0.6501 0.146 -4.4514  1e-04
constant   0.0552 0.015  3.6811  7e-04

```

The MA coefficient is significant as the p-value is very small. Hence the model is good and further residual Analysis is done for this model.

Residual Analysis

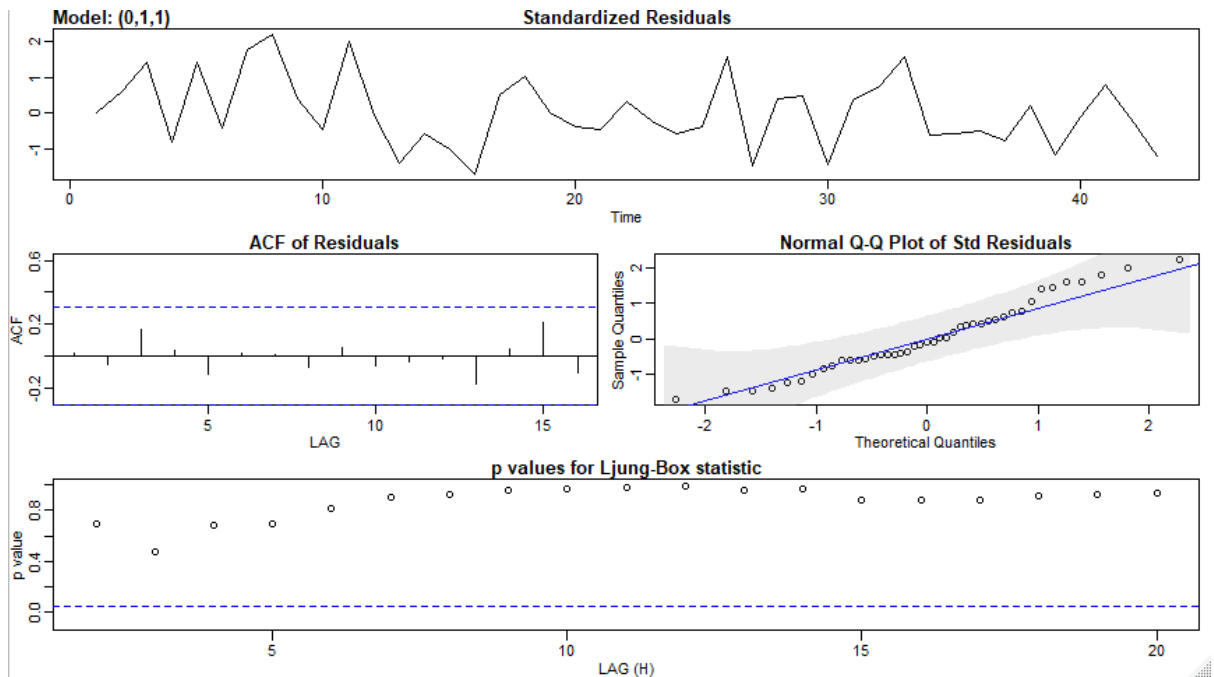


Figure 10

PACF of Residuals

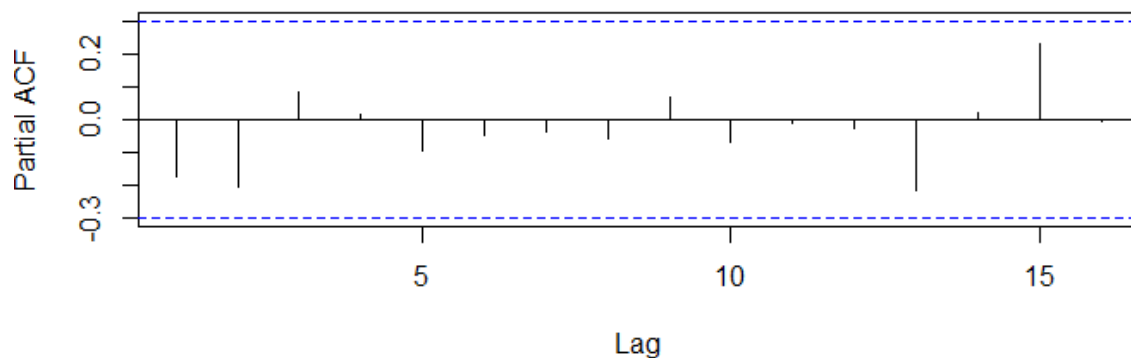


Figure 11

The standardised residuals show no specific pattern. ACF and PACF are both contained between the bounds, hence the residuals are uncorrelated. Residuals are normally distributed. The p values for all the lags for Ljung-Box statistic are high.

The ACF of the squared residuals is also contained between the bands(See Fig 12). Thus we do not consider the GARCH models here.

ARIMA(0,0,0) is the best model that can be fit to the residuals. Hence it is concluded that the residuals are white noise.

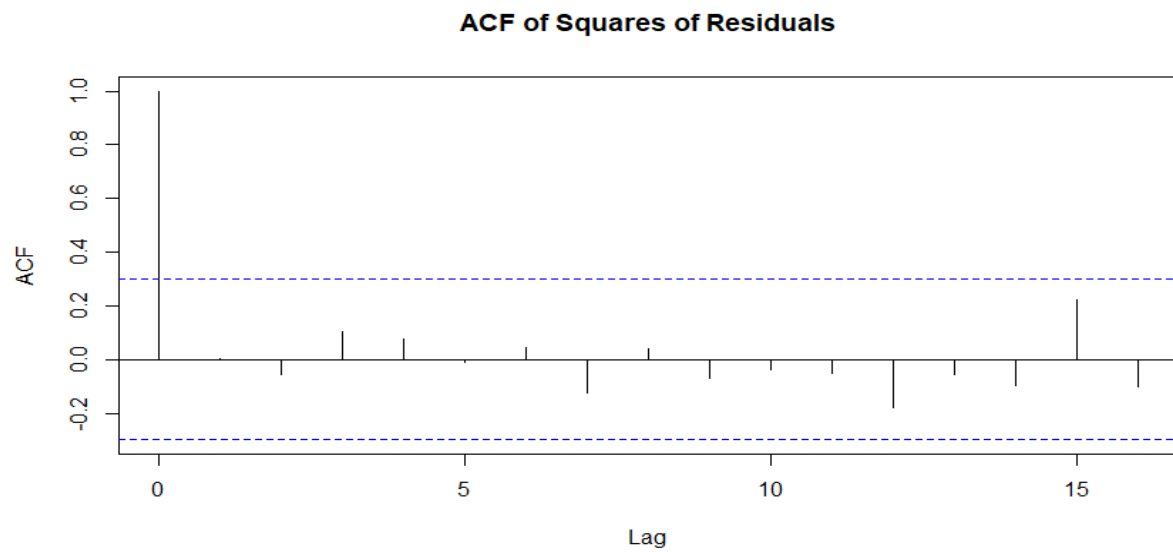


Figure 12

Formals tests carried out on the residuals as follows.

Test	Null Hypothesis	p value	Decision at 5% los
Box-Pierce test (Package : stats)	H_0 : Data independently distributed	0.253	H_0 accepted
Box-Ljung test (Package : stats)	H_0 : Data independently distributed	0.2368	H_0 accepted
Turning point test of independence (Package : spgs)	H_0 : Data not i.i.d	0.3885	H_0 accepted
Sign-Test (Package : BSDA)	H_0 : Trend not present in the series	0.3604	H_0 accepted
Mann-Kendall trend test (Package : trend)	H_0 : Trend not present in the series	0.4638	H_0 accepted

Table 2

From the above tests it is concluded that the residuals are uncorrelated and normally distributed, with no trend. Hence the **model ARIMA(0,1,1) is accepted** for the series.

FORECASTING

Forecasting on the basis of the best model ARIMA(0,1,1) is carried out. The predicted values for last 7 years along with their observed values are as follows :

Observed	Predicted
5.47	5.532988
5.71	5.532988
5.42	5.532988
5.65	5.532988
5.93	5.532988
5.81	5.532988
5.57	5.532988

Forecasts from ARIMA(0,1,1)

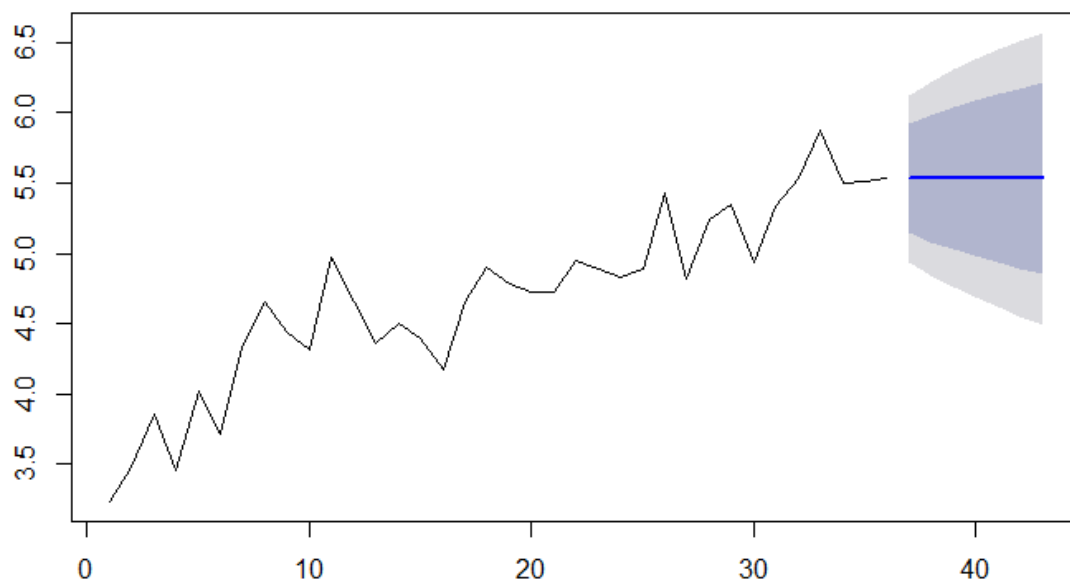


Figure 13

All the forecasts are same. They seem to be along the trend of original series.

FORECASTING using EXPONENTIAL SMOOTHING

Exponential smoothing is carried out on the series.

Parameters $\alpha = 0$, $\beta = 0$ and the damping parameter is 0.9701. The information criteria are :

Information Criteria:
AIC AICC BIC BICC
7.3407 8.3934 14.3855 16.3651

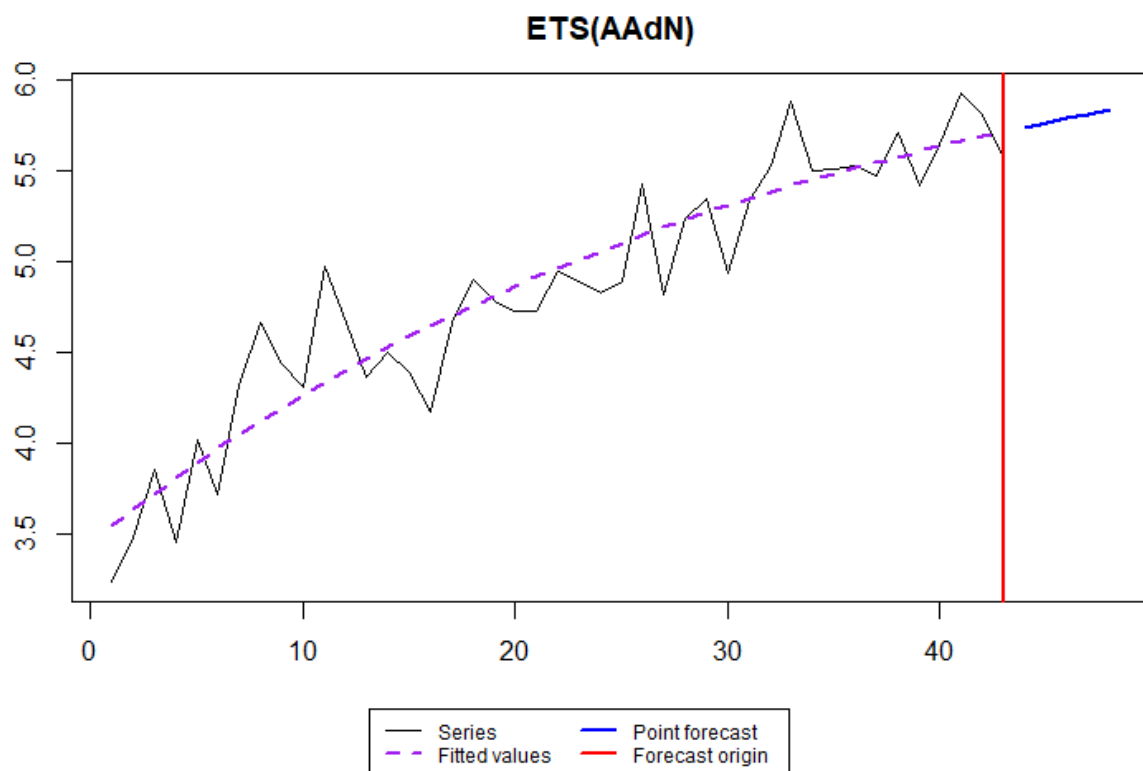


Figure 14

The forecast seems reasonable. The only drawback of this method is that we don't get any model for inference. This is useful only if our goal is forecasting.

CONCLUDING REMARKS

In this analysis, we tried to analyze the batting patterns in ODI cricket through Run Rate per Over. The average run rate has increased with the time. The presence of trend in the time series of the run rate negates the possibility of any sudden changes brought about by the introduction of newer forms of the game in the average run rate of ODI cricket. There has been a gradual rise in the average run rate. The factors which have caused the average run rate in ODI cricket to rise are pure innovations in the game-the way it is played, the kind of pitches that are used, the technological changes that have occurred and increase in competitiveness in the game.

We have found that MA(1) is the best model for RPO. All the residual assumptions are satisfied. Forecasts have been made but they are not so reasonable. Also, Exponential smoothing is used for the forecasting purpose. These forecasts are along the trend. For more accurate forecasts some advanced models can be fit.